

Keerat Singh Bhasin

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EDUCATION

University of Illinois Urbana-Champaign

B.S. Systems Engineering & Design; Minor in Computer Science; GPA: 3.7/4.0

Champaign, IL

Expected May 2027

EXPERIENCE

Neutrinos

AI Platform Enablement Intern

Remote

Mar. 2026 – May 2026

- Benchmarked 5 low-code platforms: Appian, Mendix, Unqork, Pega, and EvoluteIQ
- Built an 8-dimension evaluation framework across UX, certification architecture, and monetization
- Modeled freemium tier structure balancing conversion incentives and certification revenue
- Synthesized findings into platform strategy presented to product and engineering leadership

Engineers Without Borders UIUC

External Vice President

Champaign, IL

Nov. 2024 – Jan. 2026

- Raised \$50,000 in 2025 for infrastructure projects, up 20% year over year
- Built corporate, alumni, and donor pipeline for EWB project funding and sponsorships

Institute of Industrial and Systems Engineers

Mentorship Director

Champaign, IL

Apr. 2024 – May 2025

- Scaled mentorship program to 50+ students, alumni, and faculty across ISE
- Increased participation by 350% through structured matching and engagement strategy

RESEARCH

Product Design Lab

Undergraduate Research Assistant

Champaign, IL

Sept. 2024 – Dec. 2024

- Reverse-engineered a 1966 Ford GT40 Mk II via dual-pass 3D scanning of a 1:18 scale model in Geomagic Design X
- Processed and repaired mesh topology to produce a watertight digital twin of the original bodywork
- Redesigned exterior to FIA LMH homologation limits: 2000 mm width, 5000 mm length, 3100 mm wheelbase
- Modeled diffuser, sidepods, NACA airfoils, floor, and rear wing in Siemens NX
- Defined underbody aero surfaces under FIA LMH dimensional and downforce constraints
- Presented at UIUC Undergraduate Research Symposium and ISE Student Conference

PROJECTS

Autonomous Ball-Sorting Car | C, TI C2000, OpenMV, LiDAR, AprilTags, PID Control, SOLIDWORKS

- Built an autonomous car that drove, detected, classified, collected, and delivered colored balls
- Designed a ball collection and release mechanism in SOLIDWORKS, integrating a servo kicker for actuation
- Implemented FSM-based behavior with PID control, dead reckoning, and waypoint navigation
- Used LiDAR wall-following and timed search turns for arena exploration and obstacle avoidance
- Integrated OpenMV LAB color thresholds and AprilTags for ball classification and delivery

3D-Printed Suitcase Handle FEA Optimization | Abaqus, FEA, Design Optimization, 3D-Printed Nylon

- Optimized a 3D-printed nylon handle in Abaqus for 100 lb distributed loading
- Kept final ergonomic design at 0.292 lb, over 51% below the 0.6 lb mass limit, with FOS ≥ 7
- Reduced stress concentrations using fillets, thickness changes, and geometry edits

TECHNICAL SKILLS

Engineering: GD&T, DFM, FEA, rapid prototyping, testing, 3D printing, machining, structural analysis

Controls: PID control, embedded systems, Simulink, LabVIEW, LiDAR/camera perception, UART, SPI, JTAG

CAD/CAE: Siemens NX, Fusion 360, SOLIDWORKS, Geomagic Design X, Abaqus, Onshape

Programming: Python, Java, C++, C, MATLAB

Certifications: Six Sigma Green Belt, Lean Green Belt